TECHNICAL ARCHITECTURE

MediaPipe in seeneyu

How seeneyu Delivers Real-Time Body Language Analysis
Client-Side ML Pipeline · Zero API Cost · Instant Feedback

VERSION
1.0 — April 2026

STATUS
Production

BRANCH
master

MODELS
FaceLandmarker + PoseLandmarker

MIGRATION
Commit b56564b (2026-03-25)

LIVE
seeneyu.vercel.app

Table of Contents

MediaPipe Technical Architecture · April 3, 2026 · 9 Chapters · 13 Source Files

01 Executive Summary — What MediaPipe is and why seeneyu uses it
02 Architecture Overview — Client-side ML pipeline and data flow
03 Core Infrastructure — Module-by-module breakdown of the ML pipeline
04 Features Using MediaPipe — Three recording modes powered by MediaPipe
05 Scoring Engine — Deep dive into expression-scorer.ts
06 Feedback Generation — Dual-path coaching feedback system
07 The Migration — GPT-4o Vision → MediaPipe transition
08 Pros and Cons — Honest assessment of the MediaPipe approach
09 Alignment with seeneyu Vision — How MediaPipe serves the mission

1. Executive Summary

Google MediaPipe is an open-source, cross-platform framework for building multimodal ML pipelines. In seeneyu, it powers *real-time body language analysis* — detecting facial expressions (52 ARKit blendshapes) and body posture (11 key landmarks) directly in the user's browser, with zero server-side computation.

> **Key insight:** By running ML inference client-side via WebAssembly + WebGL, seeneyu delivers instant feedback (<100ms per frame), eliminates API costs entirely, preserves user privacy (no video leaves the device), and scales to unlimited concurrent users at zero marginal cost.

MediaPipe replaced GPT-4o Vision as the primary scoring engine in March 2026 (commit `b56564b`). GPT-4o is retained only for optional rich text coaching feedback — all numerical scoring is now deterministic and reproducible.

Before vs After

Metric	GPT-4o Vision (before)	MediaPipe (after)
Latency	3-8 seconds	<100 milliseconds
Cost per scoring	$0.01-0.03	$0.00
Privacy	Video frames sent to OpenAI	All processing on-device
Reproducibility	Non-deterministic	Deterministic
Offline capable	No	Yes (after model download)
Concurrent users	Rate-limited by API	Unlimited

2. Architecture Overview

seeneyu's MediaPipe integration follows a **client-capture** → **server-score** → **feedback-generate** pipeline:

```
┌──────────────── Browser (Client) ────────────────┐
│                                                  │
│  Webcam → HTMLVideoElement → useMediaPipe hook   │
│                   │                              │
│         ┌─────────┴──────────┐                   │
│         │                    │                   │
│  FaceLandmarker        PoseLandmarker            │
│  (52 blendshapes)      (11 landmarks)            │
│         │                    │                   │
│         └─────────┬──────────┘                   │
│                   ▼                              │
│        AnalysisSnapshot[]                        │
│  (collected every 500ms/1000ms)                  │
│                   │                              │
└───────────────────┼──────────────────────────────┘
                    │  HTTP POST (JSON)
                    ▼
┌──────────────── Server (Next.js API) ─────────────┐
│                                                   │
│  expression-scorer.ts                             │
│  ├── Cosine similarity (blendshape matching)      │
│  ├── Vector magnitude (intensity)                 │
│  ├── Pose geometry (shoulder openness, head tilt) │
│  └── Consistency analysis (std deviation)         │
│                   │                               │
│                   ▼                               │
│  feedback-generator.ts                            │
│  ├── GPT-4o TEXT (if available) — rich coaching   │
│  └── Template fallback — zero AI dependency       │
│                                                   │
└───────────────────────────────────────────────────┘
```

Key Design Decisions

- **Client-side ML**: WASM + WebGL execution via `@mediapipe/tasks-vision`. No server GPU required.
- **Singleton loader**: Models load once per session and are cached by the browser (~8MB total).
- **GPU delegate**: Both FaceLandmarker and PoseLandmarker use WebGL GPU acceleration.
- **Snapshot-based**: Instead of continuous streaming, snapshots are collected at fixed intervals (500ms for coaching, 1000ms for arcade) to balance accuracy and performance.

- **Deterministic scoring**: All score computation uses pure math (cosine similarity, vector operations, standard deviation) — no randomness, no API calls.

3. Core Infrastructure

3.1 mediapipe-init.ts — Singleton Loader

Location: `src/lib/mediapipe-init.ts`

Lazy-loads FaceLandmarker and PoseLandmarker from Google CDN as singletons. Must only be imported dynamically from client components (WASM + WebGL requirement).

Model	CDN Source	Size	Capabilities
FaceLandmarker	storage.googleapis.com/.../face_landmarker.task	~5 MB	52 blendshapes, face detection
PoseLandmarker (lite)	storage.googleapis.com/.../pose_landmarker_lite.task	~3 MB	33 landmarks (11 key ones extracted)

Configuration:

- `delegate: 'GPU'` — WebGL acceleration
- `runningMode: 'VIDEO'` — optimized for continuous frame analysis
- `numFaces: 1` / `numPoses: 1` — single-person focus
- `minDetectionConfidence: 0.5` — balanced sensitivity

```
// Usage pattern (dynamic import from client component)
const { getAll } = await import('@/lib/mediapipe-init')
const { face, pose } = await getAll()
// face: FaceLandmarker, pose: PoseLandmarker — both ready
```

3.2 useMediaPipe.ts — React Hook

Location: `src/hooks/useMediaPipe.ts`

Client-side React hook that initializes MediaPipe and provides a `detectAll()` function for per-frame analysis.

```
const { isReady, isLoading, error, detectAll } = useMediaPipe()

// detectAll: (video: HTMLVideoElement, timestampMs: number) => AnalysisSnapshot | null
```

The hook:

1. Dynamically imports `mediapipe-init.ts` on mount
2. Runs `detectForVideo()` on both FaceLandmarker and PoseLandmarker
3. Extracts 52 blendshape values from face result
4. Extracts 11 key pose landmarks using MediaPipe standard indices
5. Returns an `AnalysisSnapshot` combining both results

Pose Landmark Indices (MediaPipe Standard)

Landmark	Index	Landmark	Index
Nose	0	Left Shoulder	11
Left Ear	7	Right Shoulder	12
Right Ear	8	Left Elbow	13
Left Wrist	15	Right Elbow	14
Right Wrist	16	Left Hip	23
		Right Hip	24

3.3 mediapipe-types.ts — Shared Type Definitions

Location: `src/lib/mediapipe-types.ts`

Defines TypeScript interfaces used by both client and server code:

Interface	Purpose	Key Fields
`PoseLandmarkData`	Extracted pose positions	11 landmarks × {x, y, z} (normalized 0-1)
`AnalysisSnapshot`	Single analysis frame	timestampMs, blendshapes (52 values), poseLandmarks, faceDetected
`ArcadeAnalysisPayload`	Client → server for Arcade	challengeId, snapshots[], peakSnapshot
`MicroAnalysisPayload`	Client → server for Micro-Practice	snapshots[]

3.4 analysis-helpers.ts — Snapshot Collection

Location: `src/lib/analysis-helpers.ts`

Provides the `startAnalysisCollection()` factory that creates an `AnalysisCollector` — collecting MediaPipe snapshots at regular intervals during video capture.

```
const collector = startAnalysisCollection(videoElement, detectAll, 500)
// ... user records for 10-60 seconds ...
const snapshots = collector.stop()      // AnalysisSnapshot[]
const peak = collector.getPeakSnapshot() // highest expression intensity
```

Peak snapshot selection: Computes overall expression intensity as the sum of all non-neutral blendshape values. The snapshot with highest total intensity becomes the "peak" — used as the primary scoring input for Arcade challenges.

4. Features Using MediaPipe

MediaPipe powers all three recording/practice modes in seeneyu:

4.1 Arcade Mode

Quick 10-second expression challenges.

- **Component:** `src/app/arcade/[bundleId]/page.tsx`
- **Collection interval:** 1000ms (1 snapshot/second)
- **Scoring:** Peak snapshot + all snapshots → `scoreArcadeAttemptFromAnalysis()`
- **Output:** 0-100 score with breakdown (expression_match, intensity, context_fit)
- **API cost:** $0.00

4.2 Micro-Practice

Duolingo-style step-by-step skill practice with 30-second recordings.

- **Component:** `src/components/MicroPracticeFlow.tsx` + `PracticeRecorder.tsx`
- **Collection interval:** 500ms (2 snapshots/second)
- **Scoring:** Skill-specific blendshape analysis → `scoreMicroPracticeFromAnalysis()`
- **Output:** Pass/needs-work verdict, 4 dimension scores, positives, improvements, actionable tips
- **API cost:** $0.00

4.3 Full Performance

Complete 30-60 second recording mimicking a movie clip.

- **Component:** `src/app/library/[clipId]/record/RecordClient.tsx`
- **Collection interval:** 500ms
- **Scoring:** Multi-dimension analysis → `scoreFullPerformanceFromAnalysis()`
- **Feedback:** GPT-4o TEXT coaching (optional) OR template-based (zero AI) via `feedback-generator.ts`
- **Output:** Overall score, 4 skill dimensions (1-10 each), expression/pose timelines, rich coaching feedback
- **API cost:** $0.00 for scoring; optional $0.01 for GPT-4o text coaching

Feature Comparison

Feature	Duration	Interval	Scoring Function	AI Text
Arcade	10s	1000ms	scoreArcadeAttemptFromAnalysis	No
Micro-Practice	30s	500ms	scoreMicroPracticeFromAnalysis	No

Feature	Duration	Interval	Scoring Function	AI Text
Full Performance	30-60s	500ms	scoreFullPerformanceFromAnalysis	Optional

5. Scoring Engine

Location: `src/services/expression-scorer.ts` (776 lines)

The scoring engine is the heart of seeneyu's analysis pipeline. It converts raw MediaPipe data into meaningful scores through deterministic math — no AI API calls, no randomness.

5.1 Expression Pattern Library

A curated library of **25+ expression patterns** mapping human-readable expression names to expected blendshape target values (0.0-1.0):

Category	Expressions	Key Blendshapes
Basic Emotions	surprise, happy, smile, joy, anger, sad, fear, disgust, contempt	eyeWide, mouthSmile, browDown, mouthFrown, noseSneer, jawOpen
Communication	confident, stern, skeptical, empathy, warm, intense, focused	browDown, mouthPress, browInnerUp, cheekSquint, eyeSquint
Body Language	defiant, authority, calm, listening, concerned, determined	jawForward, mouthPress, browInnerUp, mouthSmile
Specific Actions	raised_eyebrow, wink, pout, speaking	browOuterUp, eyeBlink, mouthPucker, jawOpen

Each pattern is a `Record<string, number>` mapping blendshape names to target activation levels. For example:

```
surprise: {
  eyeWideLeft: 0.6, eyeWideRight: 0.6,
  browInnerUp: 0.5, browOuterUpLeft: 0.4,
  browOuterUpRight: 0.4, jawOpen: 0.4,
}
```

5.2 Keyword Matching

Challenge descriptions and contexts are matched against a priority-ordered keyword map. More specific keywords take precedence (e.g., "duchenne smile" → `joy` before "smile" → `smile`). Multiple matches are merged using max-value for each blendshape.

5.3 Cosine Similarity Scoring

The primary scoring mechanism compares the user's actual blendshape vector against the target pattern vector:

```
similarity(user, target) = (user · target) / (|user| × |target|)
```

```
Score = clamp(similarity × 100, 0, 100)
```


This measures the *direction* of the expression (which muscles activated) independently of *magnitude* (how strongly). A user who activates the correct muscles at half intensity still scores well on expression match.

5.4 Intensity Scoring

Measures how strongly the user committed to the expression:

```
intensity = |userVector| / |targetVector| × 80

// Clamped to 0–100
```

5.5 Consistency Analysis

Evaluates how steady the expression was held across all snapshots using coefficient of variation:

```
CV = standardDeviation(values) / mean(values)
consistency = clamp(100 − CV × 100, 20, 100)
```

Lower variance → higher consistency score. This rewards users who hold an expression steadily rather than flashing it briefly.

5.6 Pose Analysis

Four pose metrics computed from the 11 key landmarks:

Metric	Computation	What It Measures
Shoulder Openness	shoulderDist / headSize ratio	Open vs closed body posture
Head Tilt	atan2(earDy, earDx) in degrees	Head alignment and attentiveness
Forward Lean	shoulderMidZ - noseZ	Engagement and interest
Gesture Activity	shoulderY - wristY delta	Active hand gesturing

5.7 Final Score Composition

Arcade Mode

```
finalScore = expressionMatch × 0.5 + intensity × 0.25 + contextFit × 0.25
```

Micro-Practice

Uses skill-specific blendshape groups (e.g., eye-contact uses `eyeLookUp/Down/In/Out` + `eyeBlink`). For posture-dominant skills, pose score weighted 70% vs facial 30%.

Full Performance

4-dimension scoring (1-10 each) using skill-specific dimension labels. Overall score = mean of dimensions × 10.

6. Feedback Generation

Location: `src/services/feedback-generator.ts`

After MediaPipe scoring produces numerical metrics, the feedback generator transforms them into rich coaching advice. It uses a **dual-path system**:

6.1 AI Path (GPT-4o TEXT)

When `OPENAI_API_KEY` is available, a structured prompt sends the MediaPipe scores to GPT-4o for rich text generation:

- References specific facial muscles (orbicularis oculi, zygomaticus, frontalis)
- Compares user performance to the reference movie scene
- Generates numbered action plan steps targeting the weakest dimensions
- Provides specific exercises with hold durations and repetitions
- Cost: ~$0.01 per feedback request

Important: GPT-4o receives only numerical scores and text context — *no video or image data*. This is a TEXT-only call, not Vision.

6.2 Template Path (Zero AI)

When no API key is available (or if the AI call fails), template-based generation provides immediate feedback:

- Skill-specific positive/improvement pools for all 5 skill categories
- Dynamic template selection based on strongest/weakest dimensions
- Coaching tips library (Triangle Technique, Power Reset, Lean-In Rule, etc.)
- Action plan generated from dimension scores sorted by weakness
- Cost: $0.00

6.3 Output Structure

Field	Type	Description
overallScore	number (0-100)	Aggregate performance score
dimensions	DimensionScore[]	4 skill-specific scores (1-10)
summary	string	2-sentence performance overview
positives	string[]	What the user did well
improvements	string[]	Specific corrections with body part references
steps	ActionPlanStep[]	Numbered action items (what + why)
tips	FeedbackTip[]	Named techniques with exercises

Field	Type	Description
modelUsed	string	'mediapipe+gpt-4o-text' or 'mediapipe-local'

7. The Migration

On March 25, 2026 (commit `b56564b`), seeneyu migrated from GPT-4o Vision to MediaPipe for all recording scoring. This was one of the most impactful architectural changes in the project.

7.1 Why We Migrated

- **Cost:** GPT-4o Vision charged $0.01-0.03 per scoring request. At scale (1000+ daily users × 5 practices), this would cost $50-150/day.
- **Latency:** API round-trips added 3-8 seconds of waiting. Users expected instant feedback during practice.
- **Privacy:** Video frames were being sent to OpenAI's servers. Some users (enterprise, education) require data to stay on-device.
- **Reproducibility:** GPT-4o Vision gave different scores for the same input. This made progress tracking unreliable.
- **Availability:** API outages or rate limits disrupted the entire coaching flow.

7.2 What Changed

New Files (6)

File	Purpose	Lines
src/lib/mediapipe-init.ts	Singleton model loader	70
src/lib/mediapipe-types.ts	Shared TypeScript interfaces	43
src/lib/analysis-helpers.ts	Snapshot collection during recording	77
src/hooks/useMediaPipe.ts	React hook for detection	135
src/services/expression-scorer.ts	Deterministic scoring engine	776
src/services/feedback-generator.ts	Coaching text generation	243

Modified Files (8)

All three recording flows (Arcade, Micro-Practice, Full Performance) were updated to use the `useMediaPipe()` hook and `startAnalysisCollection()` helper. Server-side API routes switched from GPT-4o Vision calls to the new scoring functions.

7.3 What Was Preserved

- **GPT-4o TEXT** retained for optional rich coaching feedback (not Vision — text only)
- **Template fallback** ensures feedback works even without any API key
- **Recording upload** to Vercel Blob still works for video playback/review

- **All existing UI** unchanged — the migration was invisible to users

8. Pros and Cons

8.1 Advantages

#	Advantage	Impact
1	**Zero API cost for scoring**	All ML inference runs client-side. No per-request charges. Cost savings scale linearly with user growth.
2	**Instant feedback (<100ms)**	No network round-trip for scoring. Users get immediate results after recording. Feels responsive and game-like.
3	**Works offline**	After initial model download (~8MB), scoring works without internet. Critical for workshops, classrooms, and areas with poor connectivity.
4	**Deterministic and reproducible**	Same input always produces same score. Enables meaningful progress tracking over time.
5	**Privacy-preserving**	No video data leaves the user's device for scoring. Compliant with strict data protection requirements.
6	**Unlimited concurrent users**	Each user's browser does its own computation. No server-side bottleneck or rate limiting.
7	**52 blendshapes = granular analysis**	ARKit-compatible blendshapes capture fine-grained facial movements (individual eyebrows, lip corners, nose wrinkle, cheek puff).

8.2 Limitations

#	Limitation	Mitigation
1	**No audio/tone analysis**	Vocal pacing skill relies on mouth movement patterns (jawOpen, mouthFunnel) as proxy. OpenAI Whisper available for actual audio transcription in separate flow.
2	**Requires modern browser + WebGL**	Graceful error handling in useMediaPipe hook. Clear error message: "Please use a modern browser with WebGL support."
3	**~8MB model download on first visit**	Browser caches models after first download. Loading state shown during initialization.
4	**Less nuanced than GPT-4o Vision for complex social cues**	MediaPipe measures muscle activation, not intent or context. GPT-4o TEXT coaching compensates with contextual advice.
5	**Blendshape accuracy varies by lighting/angle**	Minimum confidence thresholds (0.5) filter low-quality detections. Only frames with `faceDetected: true` are scored.
6	**Pose limited to upper body**	Only 11 of 33 landmarks extracted (head to hips). Sufficient for communication coaching but not full-body dance/martial arts.

#	Limitation	Mitigation
7	**Single-person only**	numFaces: 1, numPoses: 1. Multi-person interaction coaching not supported.

9. Alignment with seeneyu Vision & Mission

> **seeneyu's mission:** Democratize body language coaching by making world-class communication training accessible, affordable, and private — for anyone with a browser and a webcam.

9.1 Accessible

MediaPipe runs in any modern browser — no app install, no special hardware, no GPU server. A student in rural Vietnam with a $200 laptop and a webcam can practice the same exercises as a corporate executive with a $3000 MacBook. The ~8MB model download is a one-time cost, cached by the browser.

9.2 Affordable

Zero marginal cost per scoring means seeneyu can offer generous free tiers without bleeding money. The Explorer (free) tier provides real AI-powered coaching, not a crippled demo. This aligns with the 3-tier monetization strategy: users upgrade for more content and recording time, not for basic functionality.

9.3 Private

No video data leaves the device for scoring. Users can practice embarrassing expressions, work on insecurities, and experiment freely — knowing their recordings are analyzed locally and only the numerical results reach the server. This is especially important for:

- **Education:** Student privacy regulations (FERPA, GDPR)
- **Enterprise:** Corporate training where video leakage is unacceptable
- **Personal:** Users practicing vulnerable social skills

9.4 The Core Loop, Supercharged

MediaPipe transforms the Watch → Observe → Mimic → Feedback → Repeat loop from a slow, expensive cycle into a fast, free one:

Loop Step	Before (GPT-4o Vision)	After (MediaPipe)
Watch	YouTube clip	YouTube clip (unchanged)
Observe	Observation guide	Observation guide (unchanged)
Mimic	Record + upload frames	Record + analyze locally
Feedback	Wait 3-8s for API → score	Instant score (<100ms)
Repeat	$0.03 per attempt → hesitate	$0.00 per attempt → freely retry

The instant, free feedback loop encourages **more practice attempts per session**, which is the single strongest predictor of skill improvement in deliberate practice research.

9.5 Future Directions

- **Hand gesture tracking:** MediaPipe HandLandmarker (21 landmarks per hand) for gesture coaching
- **Multi-person:** Raise numFaces/numPoses for conversation practice between two people
- **Temporal patterns:** Sequence analysis (expression transitions over time, not just snapshots)
- **Audio integration:** Combine MediaPipe visual data with Whisper transcripts for holistic communication scoring
- **On-device model fine-tuning:** WebNN API (emerging) could enable personalized models that adapt to each user's baseline expressions